ANALYTICAL CHEMISTRY CUME
October 6, 2000

1. (40 points) Bi^{3+} forms a colored complex with the ligand XO (xylenol orange). The data below were obtained at 565 nm in 0.1 N H_2SO_4 with $[Bi^{3+}] + [XO] = 2.4 \times 10^{-5}$ M. When [XO] is at least 5 times $[Bi^{3+}]$, all Bi^{3+} forms complex.

$[Bi^{3+}]/([Bi^{3+}]+[XO])$	Corrected absorbance
0.0	0.04
0.1	0.21
0.2	0.39
0.3	0.57
0.4	0.71
0.5	0.76
0.6	0.68
0.7	0.56
0.8	0.37
0.9	0.19
1.0	0.00

a. Determine ε at 565 nm (molar absorptivity) for Bi^{3+}, XO, and the complex.
b. What is the molar ratio of Bi to XO in the complex? (show work).
c. Calculate the formation constant (K_f) for the Bi/XO complex.

2. (30 Points) Consider a solution of the weak acid, HA which has a dissociation constant, K_a> The molecular form of the acid, HA has a molar absorptivity of ε_{HA}, while the ionic form, A^- has a molar absorptivity of ε_{A^-}. The analytical concentration of HA (i.e. all forms of HA) is c_A. Derive an expression for the absorbance of a solution of this acid as a function of c_A, K_a, $[H^+]$, ε_{A^-}, and ε_{HA}.

3. (30 points) The complex ZnQ_2^{2-} absorbs strongly with a maximum absorption at 480 nm. Neither the chelating agent, Q^{2-}, nor Zn absorb at 480 nm. When the chelating agent is present is at least a ten-fold excess, the absorbance of the solution is dependent only upon the concentration of Zn(II) and obeys Beer's Law over a large range of concentration. A solution composed of $[Q] = 8.60 \times 10^{-3}$ M and $[Zn\} = 2.3 \times 10^{-4}$ M has an absorbance of 0.690 in a 1.00 cm cell. Under the same conditions, as solution has an absorbance of 0.540 when $[Q] = 5.00 \times 10^{-4}$ M and $[Zn] = 2.3 \times 10^{-4}$ M. Calculate K_f for the complexation process.

Analytical Cumulative Examination
November 15 2002

99	Es237 0.121s	Es238 0.97s	Es239 0.196s	Es240 5.34s	Es241 8s	Es242 11s	Es243 21s	Es244 37s	Es245 1.1m	Es246 7.7m	Es247 4.55m	Es248 27m	Es249 1.7h	Es250 8.6h	Es251 1.375d
98	Cf236 0.136s	Cf237 2.1s	Cf238 0.021s	Cf239 39s	Cf240 1.06m	Cf241 3.78m	Cf242 3.49m	Cf243 10.7m	Cf244 19.4m	Cf245 46.3m	Cf246 1.49d	Cf247 3.11h	Cf248 333.5d	Cf249 350.6y	Cf250 13.08y
97	Bk235 10.6s	Bk236 22.1s	Bk237 30.4s	Bk238 2.4m	Bk239 1.63m	Bk240 4.8m	Bk241 8.81m	Bk242 7m	Bk243 4.5h	Bk244 4.35h	Bk245 4.94d	Bk246 1.8d	Bk247 1380y	Bk248 9y	Bk249 320d
96	Cm234 35.8s	Cm235 1.27m	Cm236 2.02m	Cm237 3.98m	Cm238 2.4h	Cm239 2.9h	Cm240 27d	Cm241 32.8d	Cm242 162.8d	Cm243 29.1y	Cm244 18.1y	Cm245 8500y	Cm246 4730y	Cm247 1.56e+07y	Cm248 3.4e+05y
95	Am233 3.2m	Am234 2.32m	Am235 15m	Am236 4.4m	Am237 1.22h	Am238 1.63h	Am239 11.9h	Am240 2.12d	Am241 432.2y	Am242 141y	Am243 7370y	Am244 10.1h	Am245 2.05h	Am246 39m	Am247 23m
94	Pu232 33.1m	Pu233 20.9m	Pu234 8.8h	Pu235 25.3m	Pu236 2.858y	Pu237 45.2d	Pu238 87.74y	Pu239 2.41e+04y	Pu240 6564y	Pu241 14.35y	Pu242 3.73e+05y	Pu243 4.956h	Pu244 8.08e+07y	Pu245 10.5h	Pu246 10.84d
93	Np231 48.8m	Np232 14.7m	Np233 36.2m	Np234 4.4d	Np235 1.084y	Np236 1.54e+05y	Np237 2.14e+06y	Np238 2.117d	Np239 2.356d	Np240 1.032h	Np241 13.9m	Np242 5.5m	Np243 1.85m	Np244 2.29m	Np245 50.3s
92	U 230 20.8d	U 231 4.2d	U 232 68.9y	U 233 1.59e+05y	U 234 0.0055	U 235 0.72	U 236 2.34e+07y	U 237 6.75d	U 238 99.2745	U 239 23.45m	U 240 14.1h	U 241 18.5m	U 242 16.8m	U 243 3.18m	U 244 25.8s
91	Pa229 1.5d	Pa230 17.4d	Pa231 3.28e+04y	Pa232 1.31d	Pa233 26.97d	Pa234 6.7h	Pa235 24.5m	Pa236 9.1m	Pa237 8.7m	Pa238 2.3m	Pa239 1.77h	Pa240 26.6s	Pa241 17.3s	Pa242 11.4s	Pa243 4.22s
90	Th228 1.913y	Th229 7880y	Th230 7.54e+04y	Th231 1.063d	Th232 100	Th233 21.83m	Th234 24.1d	Th235 7.1m	Th236 37.5m	Th237 4.7m	Th238 9.4m	Th239 33.1s	Th240 11.2s	Th241 8.17s	Th242 2.32s
89	Ac227 21.77y	Ac228 6.15h	Ac229 1.05h	Ac230 2.03m	Ac231 7.5m	Ac232 1.98m	Ac233 2.42m	Ac234 44s	Ac235 18.1s	Ac236 5.55s	Ac237 7.57s	Ac238 3.03s	Ac239 1.6s	Ac240 1.44s	
88	Ra226 1600y	Ra227 42.2m	Ra228 5.75y	Ra229 4m	Ra230 1.55h	Ra231 1.72m	Ra232 4.2m	Ra233 30s	Ra234 30s	Ra235 9.65s	Ra236 7.18s	Ra237 4.17s			
87	Fr225 4m	Fr226 49s	Fr227 2.47m	Fr228 39s	Fr229 50.2s	Fr230 19.1s	Fr231 17.5s	Fr232 5s	Fr233 3.92s	Fr234 2.01s	Fr235 1.88s				
86	Rn224 1.78h	Rn225 4.5m	Rn226 7.4m	Rn227 22.5s	Rn228 1.08m	Rn229 14.3s	Rn230 6.66s	Rn231 5.41s	Rn232 3.43s						
85	At223 50s	At224 10.7s	At225 16.2s	At226 6.42s	At227 7.15s	At228 5.56s	At229 2.63s								
84	Po222 9.3m	Po223 2.87m	Po224 1.56m	Po225 20.4s	Po226 10.2s										
83	Bi221 5.42s	Bi222 2.09s	Bi223 2.24s	Bi224 0.993s											